PPP-P-16D
12 SEPTEMBER 1989
SUPERSEDING
PPP-P-16C
June 15, 1967

FEDERAL SPECIFICATION

PACKAGING OF PHOTOGRAPHIC DRY AND LIQUID CHEMICALS,

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers preservation, packing, and marking requirements for dry and liquid photographic chemicals.

1.2 Classification.

1.2.1 Types and classes. Unit containers shall be of the following types and classes, as specified (see 6.2).

Type I - Polyethylene bottles.

Type II - Metal containers.

Class 1 - Cans.
Class 2 - Pails.
Class 3 - Drums.

Type III - Fiberboard drums.

Type IV - Polyethylene bags.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378	-	Plastic Sheet and Strip, Thin Gage, Polyolefin.
PPP-B-576	-	Boxes, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-569	-	Containers, Plastic, Molded (for liquids, Pastes and Powders) Overracked.

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PPP-D-723	-	Drums, Fiber
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208-Liters)
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

Federal Standards

Fed. Test Method Std. No. 151	-	Metals, Test Methods.
Fed. Std. No. 123	-	Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specification, standards and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402).

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Washington, D.C.; Atlanta; Chicago; Kansas City, MO; Fort Worth; Denver; San Francisco; Los Angeles; and Seattle, WA).

(Federal Government activities may obtain copies of Federal specification documents and the Index of Federal Specification, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-P-116	-	Preservation - Packaging, Methods of
MIL-B-26701	-	Bottles, Screw Cap and Carboys Polyethylene Plastic.

Military Standards

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

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3.3.3 Labeling of the end item. The end item shall be labeled as specified

CODE OF FEDERAL REGULATIONS

49 CFR 100-199, Department of Transportation Rules and Regulations for Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation Rules are a part of the Code of Federal Regulations available from the superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Materials. Requirements for materials used shall be as specified herein. The use of reclaimed materials shall be encouraged to the maximum extent possible. Unless otherwise specified, new containers shall be required.

3.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

3.2.1 Level A.

3.2.1.1 Unit containers.

3.2.1.1.1 Type I (polyethylene bottles). Liquids fixer, photographic, liquid concentrate form, to make 3.785 and 18.925 liter (1 and 5 gallon) baths shall be unit packed in polyethylene bottles conforming to the requirements of MIL-B-26701 and PPP-C-569. Bottles shall be supplied with plastic, plug-type caps in accordance with PPP-C-186, class 2, closure I.

3.2.1.1.2 Type II (metal containers).

3.2.1.1.2.1 Class 1 (cans). Metal cans shall conform to PPP-C-96, type III or type V, class 2, as specified (see 6.2). Rip strip type opening shall be used when inner parts must be removed and puncture type cans shall be used for free flowing dry chemicals. Cans shall be fabricated from 0.1134 (0.25) electrolytic tin plate [0.1134 kilograms (0.25 pounds)]. Unless otherwise specified, cans shall be coated by plan A of PPP-C-96.

3.2.1.1.2.2 Class 2 (pails). Metal pails conform to PPP-P-704, type II, class 2.

3.2.1.1.2.3 Class 3 (metal drums). Metal drums shall conform to PPP-D-729, type III.

3.2.1.1.3 Type III (fiberboard drums). Fiberboard drums shall conform to PPP-D-723, type II or III, grade A, as specified by the procuring activity.

3.2.1.1.4 Type IV (polyethylene bag). Polyethylene bags shall be constructed from polyethylene meeting L-P-378, type II minimum gage of .076 millimeters (.003 inch) or from waterproof kraft paper coated with polyethylene meeting the leakage test requirements of MIL-P-116. They shall be heat sealed. The heat sealed seams shall meet the heat seal seam requirements of MIL-P-116.

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ng markings. The sample unit for appearance, construction, workmanship, ar

3.2.1.1.5 Liners. When specified, bag liners shall be provided for type II, class 2 and 3, type III unit containers. The liners shall be fabricated of polyethylene film conforming to type I of L-P-378 and shall have a minimum gage of .076 millimeters (.003 inch). Other plastic films of equivalent protective capability and chemical compatibility may be used when previously approved or when authorized by the procuring activity. The bag liners shall be heat sealed, flat or square bottom, and of appropriate size for proper fit within the container. The heat sealed seams of the liners shall meet the heat seal seam requirements of MIL-P-116. The cross section of the bag shall be slightly greater than that of the container in which it is used. The length shall be sufficient to allow for triple (confectioner's) fold, sealing with gummed tape, or such other closure to insure no sifting of the contents. Excess air shall be forced out before making closure.

3.2.1.1.6 Volume of containers. Volume of the unit containers shall be such that not more than 90 percent nor less than 80 percent of the internal volume shall be utilized.

3.2.1.1.7 Compatibility. Parts of unit containers, interior coatings, liners, and covers which come in contact with the enclosed chemicals shall be inert and shall not react chemically or physically with the contents so as to be altered or to alter the strength, purity, or physical properties of the chemicals contained therein. Unless otherwise specified, containers shall be new and cleansed of manufacturing residues or other foreign contaminants prior to use.

3.2.1.2 Intermediate packing. Types I and II, class I containers up to and including one pint capacity, shall be afforded intermediate packaging. The intermediate container shall conform to PPP-B-636, class domestic. Each intermediate container shall be horizontally and vertically fitted with full height, half slotted style, double faced corrugated fiberboard pads, designed to effect a snug, nonshifting fit for the unit packages. A double faced corrugated fiberboard pad shall be placed between tiers. The interior packing arrangement and the quantity per intermediate container shall be in accordance with PPP-C-186 and the appendix to PPP-C-96.

3.2.2 Commercial. Preservation shall be in accordance with standard commercial practices.

3.3 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

3.3.1 Overpacking. Photographic dry and liquid chemicals unit packed in type II, class 2 or 3, or type III unit containers, require no overpacking.

3.3.2 Level A. Photographic chemicals unit packaged as specified in 3.2 shall be packed in shipping containers conforming to the following: PPP-B-576, class 2; PPP-B-585, class 3; PPP-B-591, class II. Wirebound containers conforming to PPP-B-585 shall be lined with double faced corrugated fiberboard conforming to PPP-F-320, type CF, class weather-resistent when used for packing unit packs not intermediate packing. Corrugated fiberboard pads shall be used where necessary to accomplish a snug-non-shifting load within the shipper. In so far as practicable, all shipping containers shall contain identical quantities and be of uniform size and shape. The container utilized shall be of minimum cube and tare weight consistent with the protection it

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must afford. The gross weight of each packed shipping container shall not exceed 90.72 kilograms (200 pounds). The closure, sealing, and banding (strapping) shall be in accordance with the procedures delineated.

3.3.3 Level B. Photographic chemicals unit packed as specified in 3.2 shall be packed in shipping containers conforming to domestic class of the following: PPP-B-576, PPP-B-591, PPP-B-585, PPP-B-640 or PPP-B-636. class domestic wirebound containers conforming to PPP-B-585 shall be lined with double faced corrugated fiberboard confirming to PPP-F-320, type CF, class weather-resistant when used for packing unit packs not intermediate packing. Corrugated fiberboard pads shall be used when necessary to accomplish a snug, nonshifting load within the shipper. In so far as practicable, all shipping containers shall contain identical quantities and be of uniform size and shape. The container utilized shall be of minimum cube and tare weight consistent with the protection it must afford. The gross weight of each packed shipping container shall not exceed 90.72 kilograms (200 pounds), except the fiberboard container weight limitations shall be in accordance with the special requirements of table I of PPP-P-636. The closure, sealing, and banding (strapping) shall be in accordance with the procedures delineated in the appendix of the applicable container specification.

3.3.4 Commercial. Containers shall meet Department of Transportation regulations and regulations of common carriers as applicable to the mode of transportation.

3.4 Marking.

3.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packs and shipping containers shall be marked in accordance with Fed. Std. No. 123.

3.4.2 Military agencies. In addition to any special marking required by the contract or order, interior packs and shipping containers shall be marked in accordance with MIL-STD-129.

3.4.3 Special marking. All unit containers shall be clearly marked with any special or precautionary marking required by statute, by considerations of safety, or by the characteristics of the chemicals unit packed. Photographic material subject to storage time limitations shall clearly indicate date compounded or unit packed and expiration date.

3.5 Department of Transportation regulations. Preservation, packing, and marking shall conform to Code of Federal Regulations, 49CFR, transportation of explosives and other dangerous articles, etc, whenever and wherever applicable.

3.6 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the packaging of photographic dry chemicals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except, as otherwise

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specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the same type and item submitted for acceptance and delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern. As applicable, cans of the 3.785 and 18.925 liter (1 and 5 gallon) sizes shall be salt spray tested as indicated herein. Five sample cans shall be tested with no evidence of failure.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. Examination of the end item shall be made in accordance with the classification defects and inspection levels set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers for examination in 4.2.2.1.1, 4.2.2.1.2 and 4.2.2.1.3 and in units of shipping containers for examination in 4.2.2.1.4.

4.2.2.1.1 Examination of the end item container for defects in closure, appearance, construction, workmanship, and markings. The sample unit for this examination shall be one filled primary container.

Examine	Defects
Closure	Any leakage or seepage of contents. Closure not as specified.
Appearance, construction, and Workmanship	Cracked, crushed, or any other defect affecting serviceability.
	Not in specified container.
	Not clean; any evidence of dirt, rust, dust, or foreign matter.
Markings	Marking data omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

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4.2.2.1.2 Examination of the end item container for defects in the appearance of the chemicals, space utilization of the container, and inside coating of container. The sample unit for this examination shall be one filled primary container.

Examine	Defects
Appearance of contents of container 1 (outer Container)	Not homogenous.
	Not in powder or granular form as applicable.
	Contains lumps, not free flowing, caked.
	Any discoloration other than permitted in basic specification; material contains dark spots, rust or evidence of foreign matter.
Inner container, when specified (if more than one inner container is specified, examine each container separately)	Any leakage or seepage of contents.
	Inner container not readily removable from outer container without spilling contents of outer container.
	Not of specified material.
Contents of inner container(s)	Not in powder or granular form as applicable.
	Contains lumps, not free flowing, caked.
	Any discoloration other than permitted in basic specification; material contains dark spots, rust, or evidence of foreign matter.
Space utilization (all containers)	Less than 80 percent of internal space space utilized.
	Content of container spills when opened in an upright position.
Inside coating of container 1* (outer container)	Missing, does not completely cover inner surface, flakes, chips, or not as specified.
	Any rust on inside of container.

*Contractor shall furnish a certificate of compliance that inner coating will not react physically or chemically with the contents of the container.

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4.2.2.1.3 Examination of the end item container for defects in net contents. The sample unit for this examination shall be one filled primary container. The net contents of container 1, 2, and 3 etc., of each primary container, shall not vary more than plus or minus 3 percent from the applicable weight specified in the detail specification unless otherwise specified.

4.2.2.1.4 Examination of the end item container for defects in preservation, packing, and markings. An examination shall be made to determine that the preservation, packing, and markings comply with the requirements contained within this section. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. This examination does not apply where the primary container is the shipping container. All mixing and packaging shall be performed in areas not exceeding 23.9° Celsius (75°F) and 50 percent relative humidity.

Examine	Defects
Packaging	Unit container not unit packed as specified.
	Packaging material not as specified.
Packing	Not in accordance with contract requirements.
	Arrangements or number of unit containers per shipping container not in accordance with contract requirements.
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability.
	Inadequate application of components, such as; incomplete closure of case liners, container flaps, loose or inadequate strapping, bulged or distorted containers.
Count	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

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4.2.2.1.5 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size expressed in defects per 100 units shall be as follows:

Examination Paragraph	Inspection Level	AQL
4.2.2.1.1	I	2.5
4.2.2.1.2	S-1	1.5
4.2.2.1.3	S-2	1.0
4.2.2.1.4	S-2	4.0

4.2.3 Testing of the end item. The end item shall be tested as specified in the detail specification.

4.3 Test methods.

4.3.1 Leak test. The container, metal, glass, or plastic, shall be immersed in water at a temperature of from 15.6° to 28.3°C (60° to 83°F) for 24 hours with the uppermost part of the container at a depth of .9 meters (3 feet). At the conclusion of the immersion period, the container shall be removed, dried, opened, and the interior of the container and the contents examined. Evidence of leakage shall be cause for rejection.

4.3.2 Corrosion-resistance test. The unit container shall be subjected to a salt-spray test, method 811, of Fed. Test Method Std No. 151 for 96 hours and the cyclic exposure test of MIL-P-116 following the salt-spray test. Evidence of corrosion shall be cause for rejection of the inspection lot.

4.3.3 Storage. Containers selected from each lot shall be stored in a constant temperature oven for 30 days at 43.3°C plus or minus 1.1°C (110°F plus or minus 2°F). The container shall then be removed and allowed to cool to room temperature for 2 hours. The container shall then be opened and the powders examined. Evidence of hard caking, discoloration, or moisture shall be cause for rejection.

5. PACKAGING (Not applicable to this specification)

6. NOTES

6.1 Intended use. The packaging requirements covered by this specification are intended to provide uniform requirements for the preservation, packing, and marking of dry and liquid photographic chemicals.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type of unit container.

(c) Style of unit container, when applicable.

(d) Capacity of unit container.

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(e) Type of chemical to be unit packed.

(f) Whether bag liners are required (see 3.2.1.1.5).

(g) Selection of the applicable degree of preservation and packing required. (see 3.2 and 3.3)

(h) Additional marking when required (see 3.4.1 and 3.4.2).

6.3 Bid samples. When specified, the bidder will furnish the unit containers, of the type or types to be used to fill the order, in the number specified by the procuring activity, and at no extra cost to the procuring activity.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposal methods.

MILITARY INTERESTS:

Custodians:

Air Force - 99

Review activities:

Navy - AS

User activity:

Army - SM

Preparing activity:

Air Force - 70

Project No. PACK-0791

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